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Department of Artificial Intelligence and Machine
Database Management Systems (CD252IA)

TITLE GENETIC TRAITS AND CROP PERFORMANCE		
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Software Requirement Specification (SRS) for DBMS Project

1. Introduction

1.1 Purpose

This project aims to design and implement a relational database management system that catalogs biotechnological genetic traits in various crop species and integrates phenotypic crop performance data collected from field trials. The system will store, manage, and analyze relationships between crop genetic traits, environmental/soil/climate data, and crop performance indicators (yield, disease resistance, water efficiency). It aims to provide researchers and agronomists data-driven insights into which genetic markers correlate with desirable performance outcomes, enabling informed crop selection and breeding decisions.

1.2 Scope

The system will provide a relational database for structured agricultural and genetic data and integrate with NoSQL storage for large unstructured genomic sequences or sensor data. Core capabilities:

- Store crop, genetic trait, soil, climate, and performance records.
- Support analytical queries to compute correlations between traits and performance.
- Provide import/export for bulk data (CSV, JSON) and genomic data references.
- Offer visual reports (tables, charts) and structured query results for research analysis.
- Allow role-based access control for researchers and admins.

Primary beneficiaries: university researchers, plant geneticists, agricultural data analysts, and lab instructors.

1.3 Definitions, Acronyms, and Abbreviations

- DBMS: Database Management System
- RDBMS: Relational Database Management System

- Genotype: Genetic information of crops in the form of markers or sequences
- Phenotype: Observable crop characteristics such as yield and disease resistance
- Crop Ontology: Standard vocabulary for phenotypic traits
- CRUD: Create, Read, Update, Delete operations on database entries
- Primary Key (PK): Unique identifier for a database record
- Foreign Key (FK): Attribute linking related records across tables
- AI: Artificial Intelligence
- API: Application Programming Interface
- SNP: Single Nucleotide Polymorphism (common genetic marker type)

2. Overall Description

2.1 Product Perspective

The Genetic Traits and Crop Performance DBMS operates as a standalone platform catering to agricultural researchers and breeders. It consolidates various biological and environmental data into a relational database powered by PostgreSQL. The system incorporates AI-driven phenotype prediction modules and supports NoSQL storage for unstructured genomic sequences and sensor data. The combined database environment ensures scalable and flexible data management.

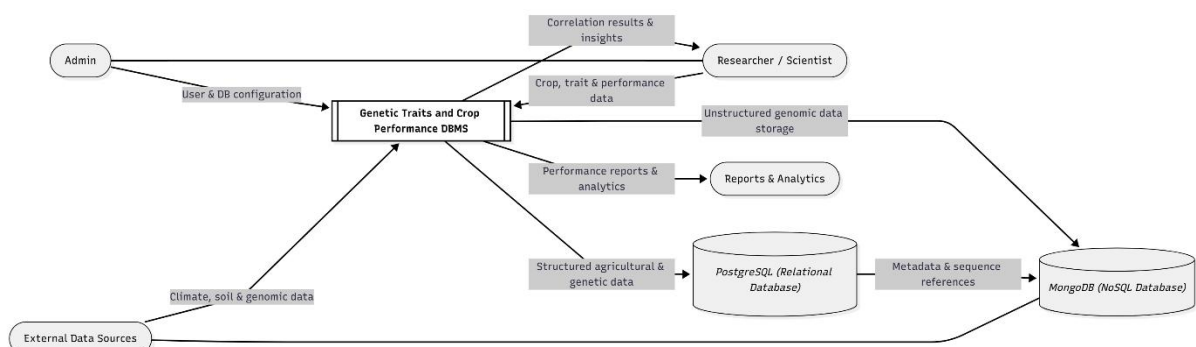


Figure 2.1: Context Diagram

2.2 Product Functions

- **User Management:** Authentication and role-based authorization (Admin, Researcher, Guest).
- **Crop Data Management:** CRUD operations for crops, varieties, and metadata.

- **Genetic Trait Catalog:** CRUD for trait definitions, gene codes, categories, and trait descriptions.
- **Performance Recording:** CRUD for yield, growth rate, disease resistance, water efficiency records linked to crops and observations.
- **Environmental Data:** Manage soil and climate records tied to regions and observation dates.
- **Research Analysis:** Store and query correlation results linking traits to performance (correlation_score, remarks).
- **Bulk Import/Export:** CSV/JSON upload and download for datasets and analysis outputs.
- **Hybrid Storage Integration:** Store large/unstructured genomic sequences or sensor logs in NoSQL (referenced from RDBMS).
- **Reporting & Queries:** Predefined analytical queries and ad-hoc query support for researchers.

2.3 User Characteristics

- **System Administrator:** Configures DB, backups, user accounts. Technical DB knowledge.
- **Researcher /Student:** Performs data entry, runs queries, views analysis reports. Familiarity with data analysis and SQL helpful.
- **Reviewer Instructor :** Reads reports, validates results, and downloads datasets.

2.4 Constraints

Hardware: development machine with minimum 8 GB RAM, recommended 16+ GB for larger genomic datasets.

Software: PostgreSQL 13+ (PostGIS optional if spatial queries needed), MongoDB 5+/6+ for NoSQL. Python 3.8+ for ETL/analysis scripts.

Data: Genomic sequences and sensor streams can be large; storage and indexing strategies required.

Privacy: If real farmer or human-subject data included, follow institutional ethics/privacy rules.

2.5 Assumptions and Dependencies

Input data (crop, trait, soil, climate, performance) is provided in consistent formats and units.

Genomic sequences stored in NoSQL are linked by identifiers (do not require full in-RDBMS storage).

Regular backups and DB maintenance will be performed by the admin.

Analyses assume standardized observation methods; temporary anomalies (unrecorded pest outbreak) may skew correlations.

3. Specific Requirements

This section defines the **functional and non-functional** requirements.

3.1 Functional Requirements

Organize requirements by modules or use cases.

Example:

Module	Requirement ID	Description
User Management	FR-1	The system shall provide user registration and secure login. Roles: Admin, Researcher, Guest.
Crop Management	FR-2	The system shall allow CRUD operations on crop records (Crop_Info).
Genetic Trait Catalog	FR-3	The system shall allow CRUD operations for genetic traits (Genetic_Traits) and support linking traits to crops.
Performance Indicators	FR-4	The system shall allow CRUD operations for performance observations (Performance_Indicators), including bulk import of observation datasets.
Soil & Climate	FR-5	The system shall store and retrieve soil and climate data (Soil_Conditions & Climate_Data) per region and date.
Research Analysis	FR-6	The system shall allow storing analysis results in Research_Analysis and retrieving them by crop, trait, date, or score range.
NoSQL Integration	FR-7	The system shall store genomic sequences and large unstructured data in a NoSQL store and link them to relational rows.

Module	Requirement ID	Description
Bulk Import/Export	FR-8	The system shall support CSV/JSON import and export of Crop_Info, Genetic_Traits, and Performance_Indicators.
Query & Reporting	FR-9	The system shall provide predefined queries: top correlated traits per crop, time-series of yield, trait vs yield plots (data exportable).
Data Validation	FR-10	The system shall validate input ranges (e.g., ph_level between 0–14; observation_date not in future) and reject invalid entries.
Audit & Logging	FR-11	The system shall log data changes (who, what, when) for critical tables.
Backup & Restore	FR-12	The system shall support scheduled backups and ability to restore critical datasets.

3.2 Non-Functional Requirements

Performance

- The system should efficiently run analytical correlation queries on datasets up to 100,000 rows with response times under 10 seconds for indexed queries.
- Bulk data imports (CSV up to 100,000 rows) must complete in a reasonable time with visible progress feedback.

Security

- User passwords must be securely hashed using bcrypt or Argon2.
- Role-based access control must restrict table deletion to Admins, and insertion of analysis results to Researchers and Admins.
- All data transfers through the web interface must be encrypted via HTTPS.

Reliability & Availability

- Daily database backups should be automatically scheduled, and restore tests must run weekly to ensure data reliability.
- The database must maintain referential integrity and transactional consistency across all operations.

Usability

- The user interface should allow simple CSV/JSON uploads and clearly display validation errors.
- Query outputs must be exportable to CSV or Excel formats for further analysis.

Maintainability

- Database schema changes should be version controlled using migration tools like Alembic.
- ETL scripts need to be modular and well-documented for easier maintenance.

Accuracy

- Correlation computations must be reproducible, with the statistical method (e.g., Pearson or Spearman) documented.
- Units for all metrics such as yield and water efficiency must be stored and presented consistently.

Scalability

- Correlation computations must be reproducible, with the statistical method (e.g., Pearson or Spearman) documented.
- Units for all metrics such as yield and water efficiency must be stored and presented consistently.